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YEAR 11 · PRELIMINARY COURSE

When petrol prices rise, do petrol thefts follow?

A research summary for **Preliminary Economics** — markets, elasticity and what happens to people's choices when the price of something changes.

Adams, N. (2026). *How do petrol thefts respond to changing prices?* Crime and Justice Bulletin No. 280. Sydney: NSW Bureau of Crime Statistics and Research.

Economics starts from a simple claim: **people respond to incentives**. Change the cost of something, or the reward for it, and behaviour changes. This study puts that claim to a hard test. It asks whether the number of people who drive off from a service station without paying — a "fail to pay" incident — goes up when the price of petrol goes up. The answer turns out to be **yes, but only a little**, and what really drives the numbers is something bigger: how far household incomes stretch.

+3%

More petrol thefts after petrol prices rise by **1%** — a real cause, not just a coincidence.

37

Extra thefts a month across NSW that a 1% price rise works out to.

1,118

Average number of petrol thefts recorded in NSW **each month in 2025**.

1 The economic thinking behind the study

In 1968 the economist Gary Becker argued that crime can be analysed like any other choice. A person weighs up the **expected benefit** against the **expected cost** — the chance of being caught, the punishment, and the **opportunity cost** of the time spent. Petrol is an unusually clean way to test this. When the price at the bowser rises, the reward from filling a tank and driving off rises automatically, while the risk of being caught does not change at all. The incentive shifts, and nothing else does.

World oil price
WTI crude, set on global markets

→

NSW petrol price
Passed through to the bowser

→

Payoff from theft rises
Same tank of fuel, higher value

→

More fail-to-pay incidents
+3% for every 1% price rise

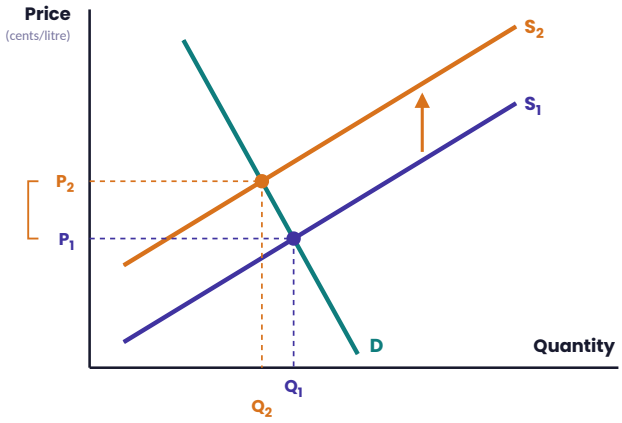
Why you can't just put the two lines on a graph

Earlier NSW research found petrol prices and service station fraud moving together almost perfectly (a correlation of **0.9**). That looks convincing — but two things can move together because a *third* thing is pushing both. A recession, a burst of inflation or population growth would lift both series at once. **Correlation is not causation**, and most of this study is about the difference.

2 Start where economists start: the market

Petrol prices are largely determined in a **market**, and the biggest influence on that market sits outside Australia altogether: the world price of crude oil. Australia buys too little oil to move that price, so a change in it is a genuine **supply shock** arriving from outside.

Figure A · A rise in world crude oil prices hits the market for petrol



1. A supply shock
A rise in the world crude oil price raises the cost of making petrol. Supply falls: the curve shifts left, from S1 to S2.

2. A new equilibrium
Price rises from P1 to P2 and quantity falls from Q1 to Q2.

3. Why the price jumps
Demand is price inelastic — people still need to drive — so most of the shock lands as a higher price, not a lower quantity. Households absorb it.

3 Everything here turns on one idea: elasticity

How much of that shock lands on the **price**, and how much on the **quantity**, is decided entirely by the shape of the demand curve — and that shape is what **price elasticity of demand (PED)** measures. There are **two ways to measure it**, and they are not rivals: they are two routes to the same answer.

Method 1 · The elasticity formula

PED = %Δ quantity demanded ÷ %Δ price

Ignoring the minus sign: a result **below 1** means demand is **inelastic**; **above 1** means **elastic**; exactly **1** is unit elastic. Precise, but you need both percentage changes to use it.

Method 2 · The total outlay method

Watch what happens to **total outlay** — the total amount consumers spend, price × quantity — when the price changes:

- Price up, outlay **up** → demand is **inelastic**
- Price up, outlay **down** → demand is **elastic**
- Outlay **unchanged** → **unit elastic**

Worked example · both methods, same data, same answer

	Before	After	Change
Price	180 c/L	198 c/L	**+10%**
Quantity demanded	500m L	485m L	**−3%**
Total outlay	$900m	$960.3m	**+$60.3m**

Formula: −3 ÷ 10 = **−0.3**. Below 1, so demand is inelastic.

Total outlay: price rose and spending rose from $900m to $960.3m, so demand is inelastic. **Same verdict.**

Hypothetical figures, chosen to illustrate the method.

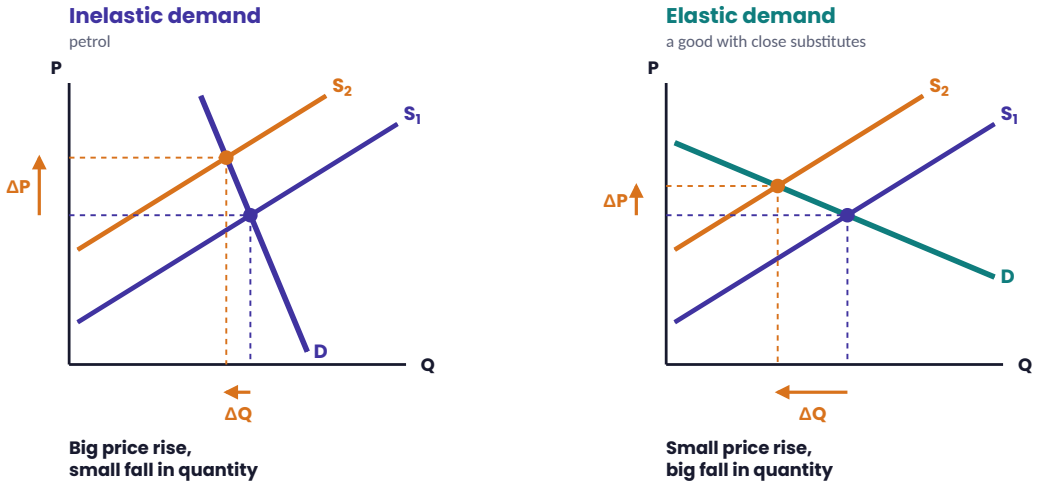
Why petrol sits at the inelastic end

- **It is a necessity.** Most driving is commuting, work or care — not easily abandoned because the price moved.
- **Few close substitutes.** Public transport is not everywhere, and nothing else fuels your car.
- **Short time period.** You cannot change your car, job or commute this week.
- **But a real share of income.** Fuel takes a large slice of a lower-income budget, so the squeeze is felt anyway.

The consequence that matters for this study

The total outlay result is the cost-of-living squeeze in one number. Households spent **$60 million more to buy less petrol** — and that money has to come from somewhere else in the budget. Inelastic demand is precisely why a fuel price shock lands on household spending rather than on quantity, and why it becomes a **cost-of-living** problem. Cost of living, as the findings on the next pages show, is what the level of petrol theft really tracks.

Figure B · The same supply shock, two different demand curves



Both panels start from the same equilibrium and apply an identical shift from S_1 to S_2. The only difference is the elasticity of demand — and it decides who absorbs the shock.

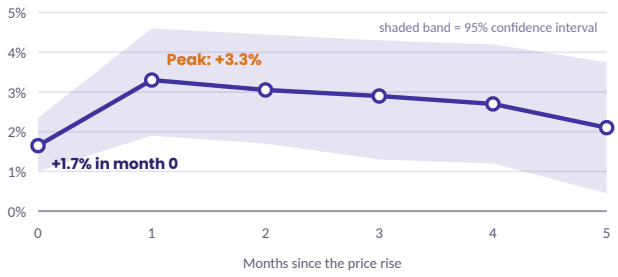
Every headline number in this study is an elasticity

Elasticity compares the % change in one thing with the % change in another. Once you notice that, the study's results line up as one ranking.

- **Theft is elastic with respect to the petrol price.** A 1% price rise causes about a 3% rise in thefts — an elasticity of roughly 3. Demand for petrol barely responds to price; *theft* responds sharply.
- **Time changes elasticity.** The response is zero on the day and builds over months — exactly as with PED.
- **And petrol is not the big one.** Figure D below compares these elasticities directly.

4 What the study actually found

Figure C · Thefts after a 1% rise in petrol prices



Cumulative % change in monthly petrol thefts, adapted from Figure 6 of the bulletin. The shaded band is the researcher's confidence interval.

Prices do cause thefts — a bit. Using world oil price movements to isolate a genuine price shock, thefts rise about 1.7% straight away and around 3.3% in total. Across NSW that is roughly **37 extra incidents a month**.

But not on the day. Comparing prices between 111 local government areas day by day found *no* measurable effect. Service stations keep their prices close to their competitors, so there is very little difference between areas to react to — and stealing petrol seems to involve some planning rather than a spur-of-the-moment choice at the pump.

Thefts fell during COVID-19 lockdowns, when people were driving far less. That is a **shift** in demand for petrol, not a movement along the curve — and it took theft down with it.

5 The bigger finding: it isn't really about petrol

Figure D · What the level of theft follows



% change in theft when each factor is 1% higher — an elasticity for each. **Patterns over time, not proven causes.**

Looking at what the **level** of theft has followed since 2016, the petrol price is the smallest part of the story. The general **cost of living** matters far more, and **wages** most of all — in the opposite direction.

Together they describe **real income**: what a household's pay actually buys. When prices climb faster than wages, purchasing power falls and budget pressure rises.

6 Where this fits the Preliminary course

References are to the **Economics Stage 6 Syllabus (2009)**, the syllabus currently taught in Year 11. The **Economics 11–12 Syllabus (2025)** begins in Year 11 from 2027; this material maps forward to **Markets**, the **Household and business sector** and the **Government sector**.

TOPIC 3 Markets

- **Price elasticity of demand — concept and measurement.** Figure B is the core diagram: identical supply shocks, opposite outcomes, because PED differs.
- **Factors affecting elasticity of demand.** All four appear in one good: necessity, few substitutes, the short time period, and the share of income fuel absorbs.
- **The total outlay method.** Price rose and total outlay rose, confirming inelastic demand — and showing exactly how a fuel shock turns into cost-of-living pressure.
- **Market equilibrium and the price mechanism.** Figure A is a standard supply shift: a higher input cost reduces supply, raising price and lowering quantity. FuelCheck, which publishes every station's price daily, exists to make that price signal visible.

TOPIC 1 Introduction to Economics

- **The economic problem, choice and opportunity cost.** Becker's model is opportunity cost applied somewhere unexpected: the higher a person's wage, the more they give up by spending time on crime.
- **The business cycle.** Research cited in the bulletin finds income-motivated crime rises during recessions, when incomes and employment fall.

TOPIC 2 Consumers and Business

- **Factors influencing consumer choice.** Income, price and the price of substitutes all appear — including driving to a cheaper LGA rather than paying more locally, a substitution response showing demand is not *perfectly* inelastic.
- **Business decision-making.** Service stations price against their competitors, not independently — a near-identical product with highly informed buyers.

TOPIC 4 Labour Markets

- **Nominal and real wages.** The largest result in the study is the wages figure. Read next to living costs, it is a statement about **purchasing power**.

TOPIC 6 Government and the Economy

- **The role of government. Regulating** the market (FuelCheck), **providing** policing, and **redistributing** income under cost-of-living pressure. Inelastic demand also makes fuel a reliable target for indirect taxation.

7 Practice questions

Use pages 1–3 of this summary as your stimulus material.

Short answer

1. **Define** price elasticity of demand, and **state** the formula used to measure it. — **3 marks**
2. **Explain** TWO factors that make the demand for petrol price inelastic. — **4 marks**
3. **Outline** ONE advantage of the total outlay method over the elasticity formula. — **3 marks**
4. Using the worked example on page 2: **(a) calculate** the PED; **(b) apply** the total outlay method; **(c) explain** why both reach the same verdict. — **5 marks**
5. With reference to Figure B, **explain** why an identical supply shock produces a much larger price rise when demand is inelastic than when it is elastic. — **5 marks**
6. **Explain** what the total outlay method reveals about petrol when its price rises, and **outline** why this matters for the cost of living. — **4 marks**
7. **Explain** why the price elasticity of demand for petrol is likely to be greater over a longer time period. — **3 marks**
8. With reference to Figure A, **explain** what happens in the market for petrol when the world price of crude oil rises. — **4 marks**
9. **Define** opportunity cost, and **explain** how it applies to the decision to steal petrol. — **3 marks**
10. The study finds that a 1% rise in petrol prices causes a 3% rise in thefts. **Explain** what this tells you about how responsive theft is to price, compared with how responsive the quantity of petrol demanded is. — **4 marks**
11. Petrol thefts fell during COVID-19 lockdowns. **Explain** ONE economic reason why, using the distinction between a shift in demand and a movement along the demand curve. — **3 marks**

Extended response — choose ONE

QUESTION 1 · 15 MARKS

Explain price elasticity of demand and the methods used to measure it, and **analyse** its importance for consumers, businesses and government in the market for petrol. In your answer, refer to the stimulus.

QUESTION 2 · 15 MARKS

Explain how the price mechanism operates in the market for petrol, and **analyse** how changes in prices and incomes influence the decisions of consumers. In your answer, refer to the stimulus.

What a strong answer does

It uses a **labelled diagram** and quotes at least one figure. The strongest answers use elasticity as the link: demand for petrol is inelastic, so a price shock lands on household budgets rather than on quantity — and budget pressure is what mattered most.

Key terms

Price elasticity of demand — the responsiveness of quantity demanded to a change in price: $\%\Delta Q_d \div \%\Delta P$.

Inelastic demand — PED below 1: quantity demanded changes proportionally less than price.

Total outlay — total consumer spending on a good: price × quantity.

Total outlay method — judging elasticity by the direction total outlay moves when price changes: up means inelastic, down means elastic.

Supply shock — a sudden change in the cost or availability of an input that shifts the supply curve.

Causation — one variable actually producing a change in another, rather than the two simply moving together.

Source. Adams, N. (2026). *How do petrol thefts respond to changing prices?* (Crime and Justice Bulletin No. 280). Sydney: NSW BOCSAR. Data: NSW Police COPS; FuelCheck; ABS; U.S. EIA. Figure B and the worked example use illustrative teaching figures, not estimates from the bulletin.